

Winston Zhang

+61 493 040 182
winston1c9hv@gmail.com
linkedin.com/in/winston-zhang-89276b208

Final year Mechanical Engineering student with a focus on simulation-driven iterative design. Proven hands-on experience in aerodynamics, CFD, FEA and composite manufacturing for high-performance applications. Led the design and validation process of FSAE aerodynamic systems and contributed to the drag-reduction optimization process in sustainable electric aviation.

EXPERIENCE

SYDNEY MOTORSPORTS FORMULA STUDENT TEAM

Sydney, Australia

CAD | Aerodynamic Design | CFD Validation | Composite Manufacturing

Advisor

February 2025 - present

Working with the aerodynamics team to validate the full aero package using the University's wind tunnel as part of graduate thesis research. Provide technical guidance on CFD validation and composite manufacturing based on prior leadership experience within the team.

Aerodynamics Section Lead

December 2023 - February 2025

Led a team of 10 student engineers in the in-house development and manufacturing of a full aerodynamics package using SOLIDWORKS, ANSYS Fluent, laser cutting, and 3D printing. Delivered a structurally robust aero package with improved rigidity over previous years while maintaining a strict 3 kg weight target. Designed a rib and spar system for the front wing to handle unexpected loads during testing and competition. Coordinated large assemblies (>1000 parts), enforced compliance with FSAE rules, and managed timelines and task delegation.

Aerodynamics Engineer

July 2022 - December 2023

Designed and validated the front wing and undertray of the SM23 vehicle using CFD. Conducted iterative design processes using a large computing cluster to optimize aerodynamic performance. Participated in mold design and manufacturing, and performed carbon fiber layups using prepreg and conventional wet-layup techniques to produce high-performance composite components.

DOVETAIL ELECTRIC AVIATION

Sydney, Australia

CAD | CFD Validation | Reverse Engineering

Computational Fluid Dynamics Engineer (Intern)

September 2023 - January 2024

Reverse-engineered a propeller nacelle assembly by generating parametric CAD models from 3D scans in SOLIDWORKS.. Validated scanned geometry in ANSYS Fluent, optimizing cooling duct geometries to minimize drag and improve airflow efficiency. Created efficient CFD models to reduce simulation time and resource consumption.

EDUCATION

THE UNIVERSITY OF SYDNEY

Bachelor of Mechanical Engineering Honours (Fluid Mechanics)

Expected Graduation: 2025

2021 - Current

TECHNICAL SKILLS

Software

- Created mechanical designs with DFM principles in mind in SOLIDWORKS and Autodesk Inventor
- Conducted FEA Analysis using SOLIDWORKS' built in simulation packages
- Conducted various numerical simulations such as FEA, topology studies, CFD and thermal analysis through the ANSYS Suite
- Wrote MATLAB scripts for simple numerical simulations such as 2D CFD, BEM as well as various data-parsing programs.
- Proficient in Python through various calculation and simulation scripts.
- Literate in the Microsoft Office Suite

Manufacturing

- Create composite parts with prepreg and conventional wet-layup techniques.
- Create composite molds with modern techniques such as hot wire cutting, laser cutting, 3D printing and conventional techniques such as hand-sculpting foam molds.
- 3D Printing and laser cutting for rapid manufacturing.
- Extensive experience in hand tools and power tools

TECHNICAL PROJECTS

- **Homelab Server** - Functions as both a local media server and a self-hosted cloud storage solution, enabling high-speed remote access and data redundancy.
- **Robotic Arm** - A 6-axis robotic arm powered by NEMA stepper motors, with motion transmitted via belt drives to reduce backlash. The accuracy was enhanced through a custom-developed zero-backlash cycloidal gearbox tailored for precise torque and speed control.
- **Custom PC Peripherals** - Engineered a 32-gram fingertip grip mouse optimized for competitive FPS gaming, and assembled a Lily58 split ortholinear keyboard with a custom-designed case for improved ergonomics and portability.
- **UNIX** - Daily-operated dual-boot Windows/Linux setups, leveraging distros such as Arch and Ubuntu derivatives for development flexibility, customization, and deep system-level understanding.

Course result history

The details listed below are for the course Bachelor of Engineering Honours (BHENGINE-03, CRICOS: 083109M). This is a copy of your online result history and not an official certified copy of the university transcript.

Student name Winston Yueshi Zhang

Printed on: 02/Feb/2025

Student ID 510515997

Student course ID 510515997/1

Course details

Course Bachelor of Engineering Honours
Credit points obtained 186
Major/Specialisation(s) Mechanical Stream Electives (Elective)

Unit of Study Results

Year	Session	Unit of study code	Unit of study name	Mark	Grade	Credit points
2021	S1C	AERO1560	Introduction to Aerospace Engineering	80.0	DI	6
2021	S1C	ENGG1111	Integrated Engineering 1	79.0	DI	6
2021	S1C	ENGG1810	Introduction to Engineering Computing	72.0	CR	6
2021	S1C	ENGP1001	Professional Engagement Program 1A		SR	0
2021	S1C	MATH1002	Linear Algebra	59.0	PS	3
2021	S1C	MATH1021	Calculus Of One Variable	72.0	CR	3
2021	S2C	AERO1400	Intro to Aircraft Construction and Design	68.0	CR	6
2021	S2C	AMME1362	Materials 1	56.0	PS	6
2021	S2C	AMME1802	Engineering Mechanics	66.0	CR	6
2021	S2C	ENGP1002	Professional Engagement Program 1B		SR	0
2021	S2C	MATH1005	Statistical Thinking with Data	59.0	PS	3
2021	S2C	MATH1023	Multivariable Calculus and Modelling	59.0	PS	3
2022	S1C	AMME2000	Engineering Analysis	50.0	PS	6
2022	S1C	AMME2450	Design and Production	75.0	DI	6
2022	S1C	AMME2500	Engineering Dynamics	59.0	PS	6
2022	S1C	AMME2700	Instrumentation	60.0	PS	6
2022	S1C	ENGP1003	Professional Engagement Program 1C		FR	0
2022	S2C	AERO2703	Aircraft Performance and Operations	77.0	DI	6
2022	S2C	AMME2200	Introductory Thermofluids	67.0	CR	6
2022	S2C	AMME2301	Mechanics of Solids	50.0	PS	6
2022	S2C	ENGP1003	Professional Engagement Program 1C		SR	0
2022	S2C	ENGP2001	Professional Engagement Program 2A		SR	0
2022	S2C	MECH2400	Mechanical Design 1	58.0	PS	6
2023	S1C	AERO3261	Propulsion	58.0	PS	6
2023	S1C	AERO3360	Aerospace Structures 1	35.0	FA	0
2023	S1C	AERO3460	Aerospace Design 1	61.0	PS	6
2023	S1C	AMME3500	System Dynamics and Control	74.0	CR	6
2023	S1C	ENGP2002	Professional Engagement Program 2B		SR	0
2023	S2C	AMME3060	Engineering Methods	66.0	CR	6
2023	S2C	AMME5902	Manufacturing of Micro- and Nanosystems	73.0	CR	6
2023	S2C	ENGP2003	Professional Engagement Program 2C		FR	0
2023	S2C	MECH3260	Thermal Engineering and Environment	54.0	PS	6
2023	S2C	MECH3361	Mechanics of Solids 2	65.0	CR	6
2024	S1C	AMME5202	Computational Fluid Dynamics	76.0	DI	6
2024	S1C	ENGP3001	Professional Engagement Program 3A		FR	0
2024	S1C	MECH3261	Fluid Mechanics 2	59.0	PS	6
2024	S1C	MECH3362	Engineering Materials for Adv Manufacturing	61.0	PS	6
2024	S1C	MECH3660	Manufacturing Engineering	77.0	DI	6
2024	S2C	AMME4401	Industrial and Product Design Engineering	68.0	CR	6
2024	S2C	ENGG3112	Interdisciplinary Engineering	81.0	DI	6
2024	S2C	ENGP2003	Professional Engagement Program 2C		UC	0

Year	Session	Unit of study code	Unit of study name	Mark	Grade	Credit points
2024	S2C	ENGP3001	Professional Engagement Program 3A		SR	0
2024	S2C	MECH3460	Mechanical Systems Design 2	77.0	DI	6
2024	S2C	MECH5255	Air Conditioning and Refrigeration	79.0	DI	6
Credit points gained						186

Your Average Marks

Year	Session	Semester average mark	Annual average mark
2021	Semester 1	74.1	NA
2021	Semester 2	62.3	68.2
2022	Semester 1	61.0	NA
2022	Semester 2	63.0	62.0
2023	Semester 1	57.0	NA
2023	Semester 2	64.5	60.8
2024	Semester 1	68.3	NA
2024	Semester 2	76.3	72.3

End of course result history